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PRIMARY RESEARCH PAPER

Are hatchery chum salmon fry a native invader? Direct and indirect effects of stocking salmon fry on stream organisms

Koh Hasegawa · Tamihisa Ohta · Satoru Takahashi

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Abstract Native species may show invasiveness toward a recipient ecosystem through increases in abundance as a result of artificial stocking events. Salmonid species are typical examples of native invaders whose abundance is increased after stocking with hatchery fish. This study evaluated the effects of hatchery chum salmon fry on sympatric wild masu salmon fry, benthic invertebrate prey, and algae, after a single stocking event in Mamachi stream, Hokkaido, northern Japan. The results suggested that the stocked hatchery chum salmon fry decreased the foraging efficiency and growth of the wild masu salmon fry through interspecific competition, and depressed the abundance of Ephemerellidae and total grazer invertebrates (Glossosomatidae, Heptageniidae, and Baetidae) through predation. Also, the hatchery chum salmon fry may increase algal biomass through depression of grazer abundance by predation (top-down effect). These results suggested that a single release of hatchery chum salmon fry into a stream may influence the recipient stream ecosystem.

Keywords Benthic invertebrates · Competition · Masu salmon · Predation · Top-down effect

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K. Hasegawa (✉) · S. Takahashi
Hokkaido National Fisheries Research Institute, Japan Fisheries Research and Education Agency, Nakanoshima, Toyohira, Sapporo 062-0922, Japan
e-mail: hasekoh@affrc.go.jp

T. Ohta
Research Institute for Humanity and Nature, Kamigamo, Motoyama, Kita, Kyoto 603-8047, Japan

Introduction

Biological invasions through human activities are recognized as a leading disturbance of aquatic and terrestrial ecosystems, typically by causing interspecific interactions such as competition and predation between an invasive species and the sympatric species (Parker et al., 1999). Although the common notion of an invasive species is that of a nonnative species that has become established beyond its native range, Carey et al. (2012) pointed out that otherwise native species also have the potential to become invasive through extreme increases in their abundance or biomass as a result of human activity.

Stocking is a common method to increase fisheries resources (Howell et al., 1999; Hilborn, 2004), but it is also an unintentional means of creating native invaders in aquatic ecosystems (cf. Hirner & Cox, 2007). Studies on the ecological impacts of stocked fishes as native invaders are largely limited to the individual and population levels, while the effects at

the community or ecosystem levels frequently remain unclear; this is true even for species of salmonids, which provide a robust model to examine the effects of stocked fishes as native invaders (Maynard & Trial, 2014; Buoro et al., 2016).

Chum salmon *Oncorhynchus keta* (Walbaum, 1792) represent an example of native-species stocking that developed into a major fisheries management approach around the North Pacific Rim. In northern Japan, nearly two billion hatchery chum salmon are stocked annually in more than 200 streams, and this number is about four times more than what was stocked annually in the late 1960s (Kobayashi, 1980; Kitada, 2014). Chum salmon show seaward migration at their fry stage, and the peak seaward migration of stocked hatchery fry was shown to occur during the first night after stocking, while fish that remain to reside in a stream will exert predation pressure on benthic invertebrates (Hasegawa & Takahashi, 2013; Takahashi et al., 2016). Thus, where unnaturally high abundances of chum salmon fry are stocked, there is potential for effects on stream-resident organisms.

Masu salmon *Oncorhynchus masou* (Brevoort, 1856) is another native salmonid that is an important fisheries resource in northern Japan (Nagata et al., 2012). Although masu salmon fry are the most likely competitor of chum salmon fry in streams due to large habitat niche overlap (Hasegawa et al., 2011), an enclosure experiment demonstrated that the foraging efficiency and growth of wild masu salmon fry was not negatively affected by the presence of hatchery chum salmon fry (Hasegawa et al., 2014a). However, the experimental design did not consider the decrement of hatchery chum salmon fry abundance due to seaward migration or the circumstance that the abundance of the hatchery fish may exceed that of wild fish through intentional stocking. Some studies suggest that the outcome of interspecific competition between dominant and subordinate salmonid species might differ when the abundance of the subordinate exceeds that of the dominant species (Rose, 1986; Kaspersson et al., 2010). This may stem from the defense of foraging territories being too costly for dominant individuals when subordinate intruders become very abundant (Kaspersson et al., 2010). Therefore, wild masu salmon may experience negative impacts where stocking with large numbers of hatchery chum salmon occurs, such as decreased foraging efficiency and/or growth, at least while the abundance of the stocked species remains high in a given stream.

How hatchery releases of chum salmon fry affect natural benthic invertebrate communities has not yet been investigated, although several studies have described the diet of hatchery chum salmon fry after release (e.g., Hoar, 1951; Kobayashi & Ishikawa, 1964; Kobayashi & Harada, 1968). These previous studies demonstrated that the hatchery chum salmon fry usually preyed on small aquatic invertebrates, such as the Chironomidae. In addition, hatchery chum salmon fry may indirectly influence algal biomass through a top-down effect, if the hatchery fry decreases the total abundance of grazer invertebrates through predation (Pace et al., 1999).

We hypothesized that stocked chum salmon fry would 1) directly negatively affect wild masu salmon fry and benthic invertebrates through competition and predation, respectively, and 2) indirectly positively affect algal biomass through a top-down effect in a natural stream. To test these hypotheses, we conducted a field survey to periodically evaluate the foraging efficiency and growth of wild masu salmon fry, the abundance of invertebrate prey, and algal biomass after a single release of chum salmon fry in a stream in its native range. Then, we discuss the effects of hatchery chum salmon fry as a native invader in a stream ecosystem.

Materials and methods

Study sites and salmon stocking program in Mamachi stream

This study was conducted in Mamachi stream, a spring-fed tributary of the Chitose River, in Hokkaido, northern Japan (Fig. 1). The stream is meandering, with a low gradient and a homogeneous pebble substrate. Native masu salmon and nonnative brown trout *Salmo trutta* (Linnaeus, 1758) are the dominant species in the fish assemblage of the stream (Hasegawa et al., 2014b). However, brown trout fry had not yet emerged during the study period. Wild (natural spawning-origin) chum salmon fry were rare and comprised only 0.6% of the chum salmon fry captured on the first night after stocking (Hasegawa & Takahashi, 2013).

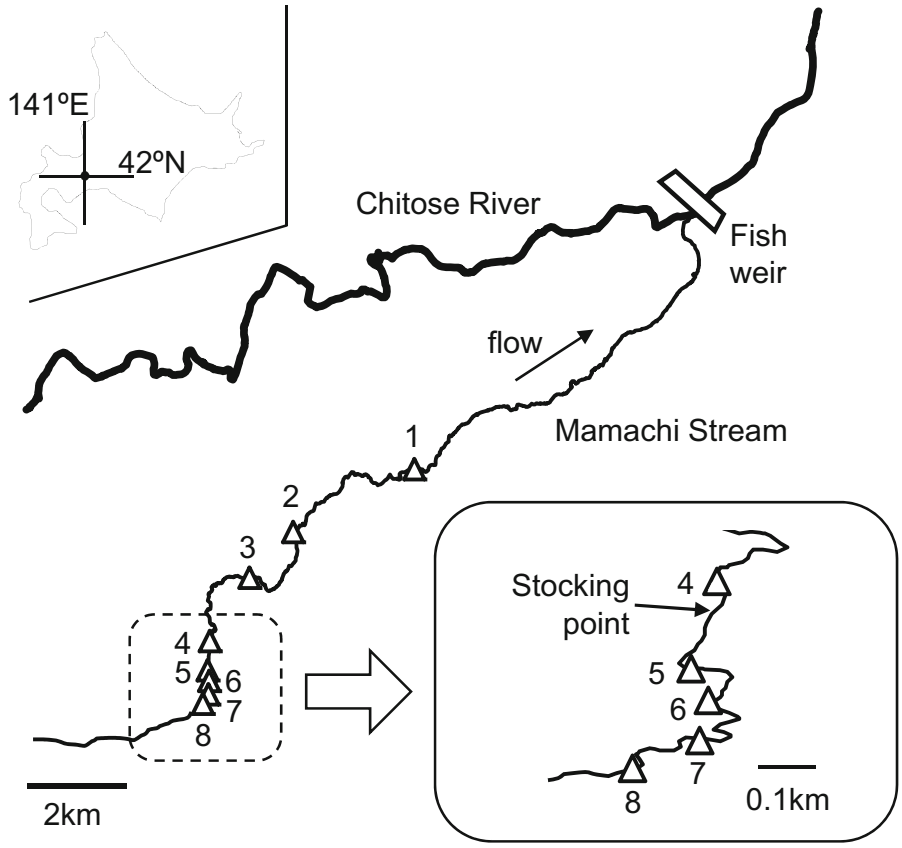

Fig. 1 Location of the study sites (triangles) in Mamachi stream, Hokkaido, northern Japan. Hatchery salmon fry were released once at the stocking point. Study sites 1–4, downstream from the stocking point, were exposed to the effects of hatchery chum salmon fry, and study sites 5–8 were not exposed. The trapezium indicates the location of a fish weir for catching adult chum salmon

Three million hatchery chum salmon fry (mean fork length: 45 mm; mean body weight: 0.7 g) were released on March 15, 2012, from the stocking point indicated in Fig. 1. The mean numbers of hatchery chum salmon fry captured using a Model-12 backpack electrofisher (Smith-Root Inc., Vancouver, WA, USA) per minute at the study sites during each sampling period were 20, 6, 5, and 2 (March 26 and April 6, 16, 25). In comparison, 1 to 5 wild masu salmon fry were captured per minute at the sites. Thus, the study sites satisfied the conditions that the abundance of hatchery chum salmon fry was greater than that of wild masu salmon fry immediately after stocking, and thereafter progressively decreased (see below for details of fish sampling).

Hatchery chum salmon fry were distinguished from wild conspecific fry by checking an otolith thermal mark using a microscope in the laboratory. Otolith thermal marking is a popular technique for identifying hatchery fry, using water temperature fluctuations to induce distinctive ring marks onto the otoliths of incubating fish (Volk et al., 1999).

Sampling of fishes, benthic invertebrates, and algae

Stocked chum salmon fry migrate downstream (Hasegawa & Takahashi, 2013), and electrofishing affirmed that no hatchery chum salmon fry were observed above the stocking point. Four study sites were chosen both below and above the single stocking point to provide sections with (study sites 1–4) and sections without (study sites 5–8) hatchery chum salmon fry (Fig. 1). Abiotic (e.g., water temperature and flow discharge) and biotic (e.g., fish fauna) conditions were similar among the study sites (e.g., Hasegawa et al., 2017), i.e., the longitudinal differences among the study sites were not clear. Fish, benthic invertebrates, and algae were sampled at the sites once before stocking (March 9, except algae sampled on March 14) and four times after stocking (March 26, and April 6, 16, 25). The survey was conducted before leaf-out of riparian trees, so light conditions in the stream were assumed to be similar among the study sites.

Forty unglazed terracotta tiles (10 × 10 cm and 0.7-cm thick) were placed at each study site as an algal culture substrate, on February 28, 2012. The tiles were positioned within a 5-m^2 area in a flat riffle. In these 5-m^2 areas, the current velocity was measured at seven evenly spaced points across three evenly spaced transects using a velocity meter (VR-201; Kenek, Tokyo, Japan). Water depth was measured at two evenly spaced points across four evenly spaced transects. Statistically significant, but small, differences in mean current velocity (range 51.4–73.0 cm/s) and mean water depth (range 16.0–25.6 cm) were detected among the eight study sites (one-way analysis of variance: current velocity: df = 7, $F = 15.70$, $P < 0.001$; water depth: df = 7, $F = 7.86$, $P < 0.001$) (Table 1).

To evaluate foraging efficiency and growth of masu salmon fry, and also to analyze diets of chum and masu salmon fry, we collected fish around the 5-m^2 areas (i.e., 25 m up and downstream from the areas) using the backpack electrofisher during all sampling periods. Fifteen wild masu salmon fry were collected from each study site, and 10 hatchery chum salmon fry were collected from study sites 1–3. These fry were euthanized with an overdose of ethyl 3-aminobenzoate methanesulfonic acid salt and immediately fixed in 70% ethanol.

In the laboratory, the body weight of each fry was measured, and the wet mass of the stomach contents was determined. Also, invertebrates found in the stomachs were separated into terrestrial and aquatic invertebrates. Aquatic invertebrates were identified to the lowest possible taxonomic level, usually genus or species, and the number of individuals in each taxon were counted in order to discuss the effect of predation on abundance of each taxon, especially grazers. The stomach-content samples of the hatchery chum salmon fry had been used by Takahashi et al. (2016) and were re-analyzed in more detail taxonomic level for this study.

To compare the abundance of benthic invertebrates between study sites with and without the hatchery chum salmon fry, five samples were collected using a Surber sampler (25 × 25 cm^2 quadrat) at each study site and during each sampling period. These samples were preserved in 70% ethanol in the field. The invertebrates were identified in the laboratory to the lowest possible taxonomic level, usually genus or species (Appendix Supplementary Material), and the number of individuals in each taxon was counted.

To compare the algal biomass between study sites with and without the hatchery chum salmon fry, the samples were collected from the eight terracotta tiles at each study site and during each sampling period. The upper surface of the tiles was brushed with running water, and the algal suspensions were bottled. The suspensions were filtered through a glass filter (No. 1822; Whatman, Maidstone, UK) in the laboratory within 24 h of sampling, and then frozen at −30°C until the chlorophyll-*a* content could be measured. The glass filters were extracted with 10 ml of 90% acetone in the dark for 24 h, and chlorophyll-*a* content in the extracts was measured using a spectrophotometer (UV-3150; Shimadzu, Kyoto, Japan).

Data analysis

Generalized linear mixed models (GLMM) were used to test the effects of hatchery chum salmon fry on stream organisms. Response variables were stomach fullness (=wet mass of stomach contents/body weight)

Table 1 Current velocity and water depth values at the study sites

Study site	Stocked hatchery chum salmon fry	Current velocity (cm/s)		Water depth (cm)	
		Mean	SD	Mean	SD
1	Presence	63.1	7.90	19.9	3.26
2	Presence	59.9	15.1	19.4	3.66
3	Presence	60.7	7.22	18.5	3.06
4	Presence	51.4	5.45	19.3	1.65
5	Absence	66.8	8.69	19.6	2.68
6	Absence	52.2	6.04	16.0	0.92
7	Absence	61.5	5.92	17.6	1.79
8	Absence	73.0	10.0	25.6	3.90

and body weight of masu salmon fry (assuming a Gaussian error distribution), abundance of the five most abundant benthic invertebrate taxa (Poisson error distribution), and chlorophyll-*a* density (Gaussian error distribution). For all GLMMs, explanatory variables were sampling date, presence/absence of hatchery chum salmon fry (study sites 1–4: presence; study sites 5–8: absence), and the interaction between sampling date and the presence/absence of hatchery chum salmon fry.

GLMMs assuming a Poisson error distribution were also performed by setting the number of prey invertebrates as the response variable, and fish species (i.e., hatchery chum salmon fry and wild masu salmon fry at all eight study sites), sampling date, and the interaction between fish species and date as explanatory variables to compare the number of individuals in the major invertebrate taxa preyed on by hatchery chum salmon fry and wild masu salmon fry.

Study site was set as a random effect in all GLMMs. The alpha level was set to 0.05. All statistical tests were performed using SPSS ver. 24 software (IBM Corp., Armonk, NY, USA).

Results

Effects of hatchery chum salmon fry on stomach fullness and growth of wild masu salmon fry

The presence of chum salmon fry was associated with reduced stomach fullness in the wild masu salmon fry, i.e., the interaction term between sampling date and presence/absence of hatchery chum salmon fry was significant (Table 2a). Estimates of stomach fullness in wild masu salmon fry were higher at the sites with hatchery chum salmon fry than at sites without chum salmon fry before stocking (Fig. 2a). After stocking, stomach fullness declined to lower levels at the study sites with hatchery chum salmon fry than at sites without chum salmon fry (March 26 and April 6) (Fig. 2a). Then, stomach fullness gradually recovered (Fig. 2a). Body weights of the wild masu salmon fry collected showed a similar, albeit less pronounced, pattern to that of estimates of stomach fullness (Fig. 2b; Table 2b).

Relationship between benthic invertebrate abundance and salmon fry diet

The differences in invertebrate abundance between study sites with and without hatchery chum salmon fry differed depending on taxa. In addition, Ephemerellidae and Glossosomatidae increased in the chum salmon fry diet during the latter half of the survey.

Chironomidae was the most abundant category of benthic invertebrate at all study sites, and sampling date affected abundance (Fig. 3a; Table 3a). The relatively higher estimates of abundance at the study sites without hatchery chum salmon fry were caused by particularly high abundance observed at study site 6. Chironomidae was also the dominant category of invertebrate in the diet of both the hatchery chum salmon fry and the wild masu salmon fry (Fig. 4a). However, the total numbers of individuals preyed on by wild masu salmon fry and hatchery chum salmon fry differed among the sampling dates (Fig. 4a; Table 4a).

Ephemerellidae was the second-most abundant category of benthic invertebrate sampled from Mamachi stream (Fig. 3b). The number of Ephemerellidae collected decreased over time, and abundance at the study site with hatchery chum salmon fry became lower than that without the hatchery chum salmon fry

Table 2 Summary of generalized linear mixed models for stomach fullness and body weight of masu salmon fry

	df1	df2	*F*	*P*
(a) Stomach fullness				
Date	4	571	22.34	<0.001
Presence/absence of hatchery chum salmon fry (P/A)	1	6	0.035	0.859
Date × P/A	4	571	8.378	<0.001
(b) Body weight				
Date	4	571	10.52	<0.001
Presence/absence of hatchery chum salmon fry (P/A)	1	6	0.131	0.730
Date × P/A	4	571	13.16	<0.001

df1 the effect degrees of freedom; *df2* the error degrees of freedom

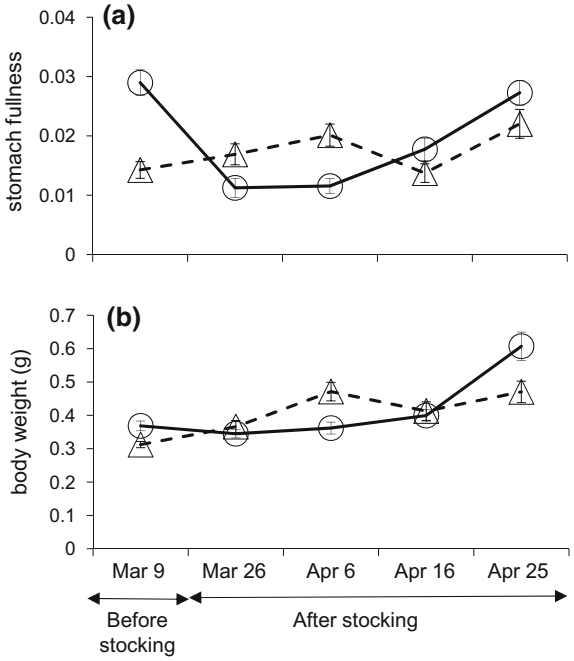

Fig. 2 Mean values (±SE) of **a** stomach fullness and **b** body weight of wild masu salmon fry sampled at the study sites with (sites 1–4) and without (sites 5–8) hatchery chum salmon fry. Sites 1–4 and 5–8 are represented by *circles* joined by *solid lines* and *triangles joined by dashed lines*, respectively

after April 6 (Fig. 3b; Table 3b). The total number of Ephemerellidae preyed on by the sampled hatchery chum salmon fry was larger than that determined for wild masu salmon fry, especially after April 6 (Fig. 4b; Table 4b).

Glossosomatidae, Heptageniidae, and Baetidae were the next most abundant invertebrate categories and constituted the major invertebrate grazers in Mamachi stream. The temporal change in the patterns of abundance differed among the taxa (Fig. 3c–e; Table 3c–e). However, the temporal changes in total abundance of the three taxa significantly differed between the study sites with and without hatchery chum salmon fry (Fig. 3f; Table 3f). The total grazer abundance at the study sites without hatchery chum salmon fry increased for March 26 to April 6, whereas abundance at the sites with hatchery chum salmon fry remained at approximately half those values (Fig. 3f). The fish stomach-contents analysis revealed that the numbers of Heptageniidae and Baetidae preyed on did not significantly differ between the fish species (Fig. 4d, e; Table 3d, e). The interaction between the sampling date and fish species was significant for the number of Glossosomatidae and total grazers consumed: that is, the numbers preyed on by hatchery chum salmon fry were apparently larger than those preyed on by wild masu salmon fry on April 16 and 25 (Fig. 4f; Table 4f).

The interaction between the sampling date and fish species was significant for the number of terrestrial invertebrates consumed (Table 4g). Terrestrial invertebrates preyed on by wild masu salmon fry, sampled on March 9 and 28, were identified as Collembola (Fig. 4g).

Comparisons of algae between study sites with and without hatchery chum salmon fry

The interaction between the sampling date and presence/absence of hatchery chum salmon fry was significant (Fig. 5; Table 5). Mean values of chlorophyll-*a* content, in algae samples from March 26 and April 6, were higher at the study sites with hatchery chum salmon fry than at sites without hatchery chum salmon fry (Fig. 5).

Discussion

Stocking a stream with hatchery chum salmon fry affected various types of sympatric aquatic organisms through interspecific competition, predation, and a top-down effect. These results suggest that the hatchery chum salmon released in Mamachi stream behaved as native invaders (cf. Carey et al., 2012), and the effects of their presence were likely to reach community or ecosystem levels in the stocked stream. Although masu salmon are widely distributed in northern Japan including Hokkaido (Watson, 1999; Nagata et al., 2012), the community structures of benthic invertebrates differ among streams even in Hokkaido (Mori et al., 2010). Therefore, the effects of hatchery chum salmon fry stocked in other streams may be somewhat different from our results.

At study sites with hatchery chum salmon fry, estimates of stomach fullness in wild masu salmon fry sharply decreased after the stocking event, and the estimates remained relatively low until April 6, but thereafter began to recover (Fig. 2a). The need for wild masu salmon fry to defend their foraging habitat from the hatchery chum salmon fry may have caused the decrease in their stomach fullness following the

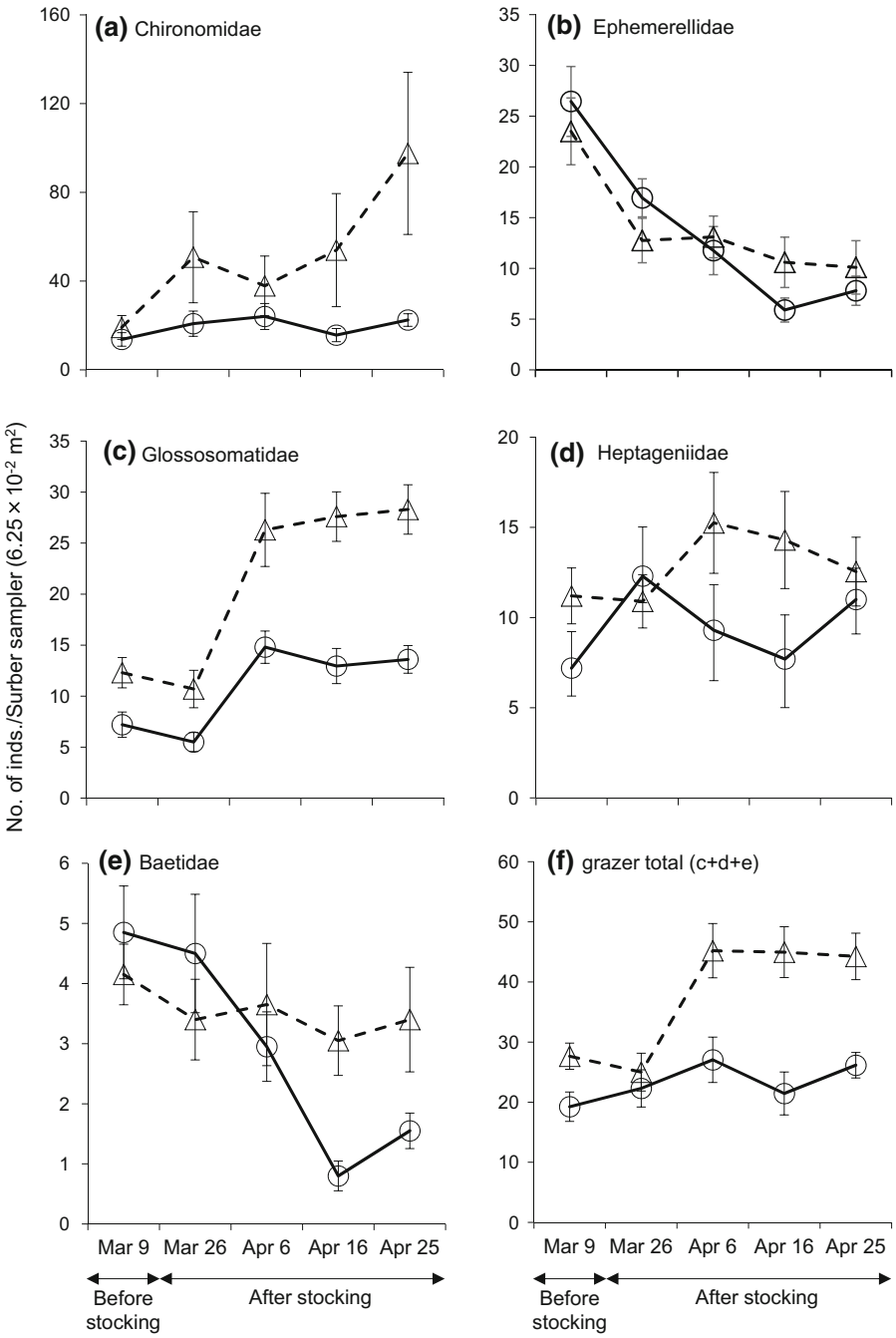

Fig. 3 Mean numbers (±SE) of total individuals of **a** Chironomidae, **b** Ephemerellidae, **c** Glossosomatidae, **d** Heptageniidae, **e** Baetidae, and **f** total number of the three major grazer taxa (**c** + **d** + **e**) collected by five Surber samples at each study site, in stream sections with (sites 1–4) and without (sites 5–8) hatchery chum salmon fry. Sites 1–4 and 5–8 are represented by *circles* joined by *solid lines* and *triangles* joined by *dashed lines*, respectively

stocking event. Although wild masu salmon fry seem to be superior competitors in relation to hatchery chum salmon fry (Hasegawa et al., 2014a), the foraging efficiency of dominant individuals may decrease with an increase in the number of subordinates. This is likely the result of more frequent encounters between the aggressive dominants and the intruders, which limits foraging time (Kaspersson et al., 2010). Therefore, the foraging efficiency of the wild masu salmon fry may have decreased, thereby decreasing stomach fullness, after the release of the hatchery chum salmon fry. However, stomach fullness of the wild masu salmon fry gradually recovered at the study sites with hatchery chum salmon fry (Fig. 2a), possibly as an outcome of

Table 3 Summary of generalized linear mixed models for densities of (a) Chironomidae, (b) Ephemerellidae, (c) Glossosomatidae, (d) Heptageniidae, (e) Baetidae, and (f) total of the three major grazers (c + d + e)

	df1	df2	*F*	*P*
(a) Chironomidae				
Date	4	190	130.3	<0.001
Presence/absence of hatchery chum salmon fry (P/A)	1	6	0.001	0.991
Date × P/A	4	190	63.20	<0.001
(b) Ephemerellidae				
Date	4	190	125.1	<0.001
Presence/absence of hatchery chum salmon fry (P/A)	1	6	0.104	0.758
Date × P/A	4	190	6.480	<0.001
(c) Glossosomatidae				
Date	4	190	85.67	<0.001
Presence/absence of hatchery chum salmon fry (P/A)	1	6	41.92	0.001
Date × P/A	4	190	1.347	0.254
(d) Heptageniidae				
Date	4	190	5.373	<0.001
Presence/absence of hatchery chum salmon fry (P/A)	1	6	0.941	0.369
Date × P/A	4	190	10.03	<0.001
(e) Baetidae				
Date	4	190	15.18	<0.001
Presence/absence of hatchery chum salmon fry (P/A)	1	6	0.341	0.580
Date × P/A	4	190	9.496	<0.001
(f) Grazers total				
Date	4	190	42.53	<0.001
Presence/absence of hatchery chum salmon fry (P/A)	1	6	6.741	0.041
Date × P/A	4	190	13.82	<0.001

df1 the effect degrees of freedom, *df2* the error degrees of freedom

the disappearance of the chum salmon fry as they migrated to the sea. Temporal changes in mean body weight, a prime growth index, roughly coincided with estimations of stomach fullness (Fig. 2b). Therefore, the presence of hatchery chum salmon fry may have affected the growth of the wild masu salmon fry through the decreasing foraging efficiency of the latter.

The effect of the hatchery chum salmon fry on benthic invertebrate abundance differed among categories. Chironomidae were the most abundant taxon sampled in the stream, and also the main prey items in stomach contents of both the hatchery chum salmon fry and wild masu salmon fry (Figs. 3a, 4a). However, the effect of hatchery chum salmon fry on Chironomidae abundance was not apparent at the study sites. The high immigration rate of Chironomidae (such as by drifting in from upstream: Elliott, 1971) may have masked the predation effect. On the other hand, the increased predation on Ephemerellidae by hatchery chum salmon fry may depress Ephemerellidae abundance after April 6 (Figs. 3b, 4b).

Estimates of total abundance of the three grazer taxa (Glossosomatidae, Heptageniidae, and Baetidae) at the study sites with hatchery chum salmon fry remained stable throughout the survey (Fig. 3c–f). Glossosomatidae had gradually colonized the experimental terracotta tiles, and their numbers increased with time; even so, Glossosomatidae abundance dropped after April 16, likely due to predation by the hatchery chum salmon fry (Fig. 4c). In contrast to Chironomidae, the abundance of Glossosomatidae may not have recovered at the study sites with hatchery chum salmon fry because of low immigration rate by these benthic invertebrates (Becker, 2001). Furthermore, added predation pressure on these three taxa from the large number of hatchery chum salmon fry may have depressed the total abundance of the invertebrate grazers. Although Heptageniidae and

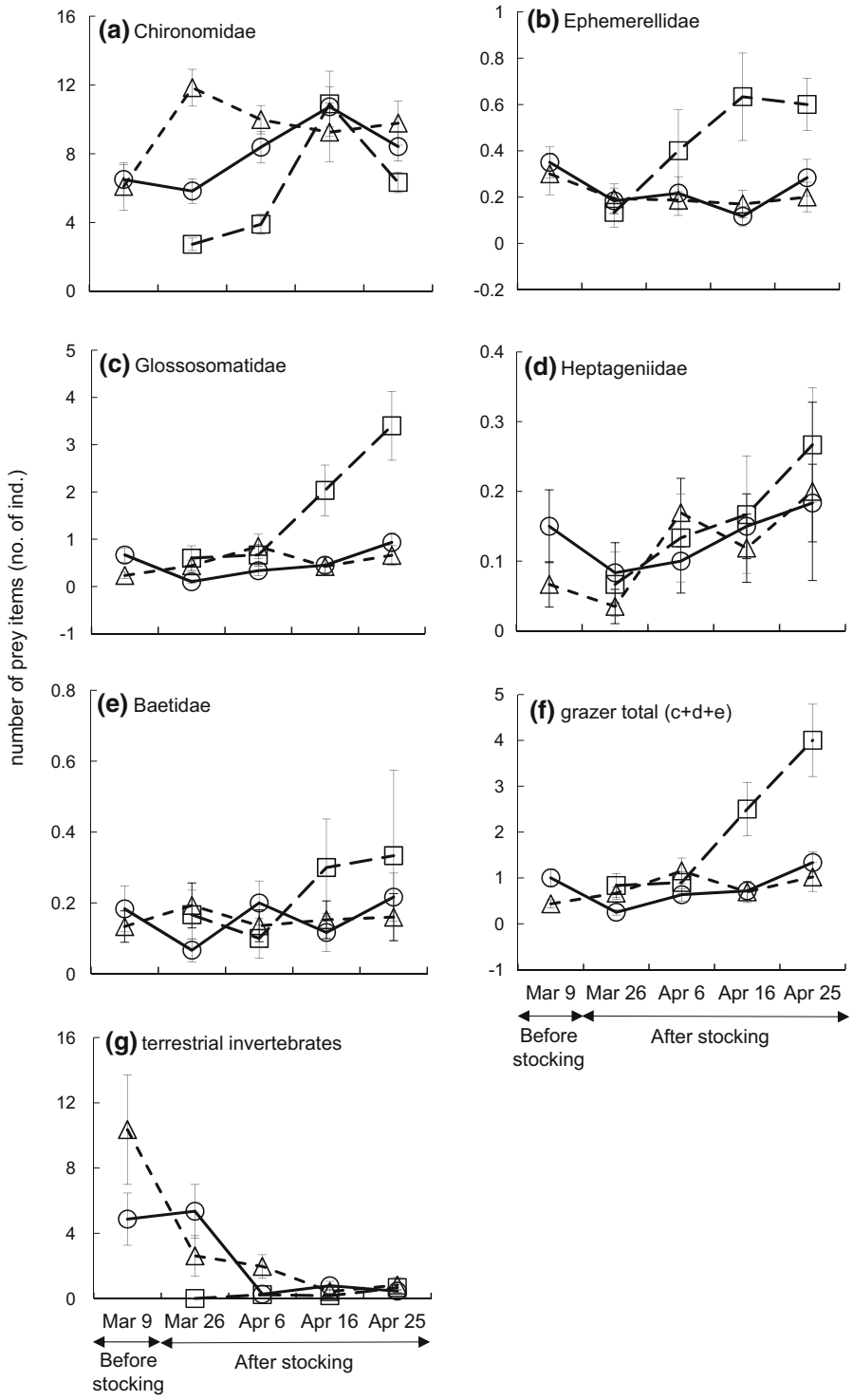

Fig. 4 Mean numbers (±SE) of prey items in each category of aquatic or terrestrial invertebrates from stomach contents of wild masu salmon fry sampled at the study sites, in stream sections with (sites 1–4) and without (sites 5–8) hatchery chum salmon fry. Masu salmon fry in site 1–4, site 5–8, and hatchery chum salmon fry are represented by *circles* joined by *solid lines*, *triangles* joined by *shorter dashed lines* and *squares* joined by *longer dashed lines*, respectively

Baetidae have high immigration rate through drifting and swimming (Elliott, 1971; Craig, 1990), a predation effect by the hatchery chum salmon fry on Baetidae abundance was likely, as Baetidae were overall less abundant than Heptageniidae in Mamachi stream (Fig. 3d, e). Thus, the supply of new

Table 4 Summary of generalized linear mixed models for the number of invertebrates in each taxon preyed on by fish

	df1	df2	*F*	*P*
(a) Chironomidae				
Date	4	691	63.79	<0.001
Fishes	2	13	57.69	<0.001
Date × fishes	7	691	33.57	<0.001
(b) Ephemerellidae				
Date	4	691	3.275	0.011
Fishes	2	13	8.348	0.003
Date × fishes	7	691	1.873	0.071
(c) Glossosomatidae				
Date	4	691	20.11	<0.001
Fishes	2	13	30.80	<0.001
Date × fishes	7	691	9.646	<0.001
(d) Heptageniidae				
Date	4	691	2.299	0.058
Fishes	2	13	0.311	0.733
Date × fishes	7	691	0.411	0.896
(e) Baetidae				
Date	4	691	0.754	0.556
Fishes	2	13	0.858	0.441
Date × fishes	7	691	0.885	0.518
(f) Grazer total (c + d + e)				
Date	4	691	18.23	<0.001
Fishes	2	13	56.45	<0.001
Date × fishes	7	691	8.066	<0.001
(g) Terrestrial invertebrates				
Date	4	691	10.45	<0.001
Fishes	2	13	0.671	0.522
Date × fishes	7	691	2.044	0.048

Fish are categorized into hatchery chum salmon fry and wild masu salmon fry at study sites with and without hatchery chum salmon fry

df1 the effect degrees of freedom; *df2* the error degrees of freedom

individuals, as mentioned above, may not have been sufficient to mask the effects of predation.

Algal biomass differed between the study sites with and without hatchery chum salmon fry, on March 26 and April 6 (Fig. 5). As chum salmon fry do not feed on algae, the additional predation pressure on invertebrate grazers by the hatchery fish may have helped determine the higher algal biomass at the study sites with chum salmon fry through a top-down effect. In addition, feces of hatchery chum salmon fry may have also contributed to increased algal biomass (bottom-up effect).

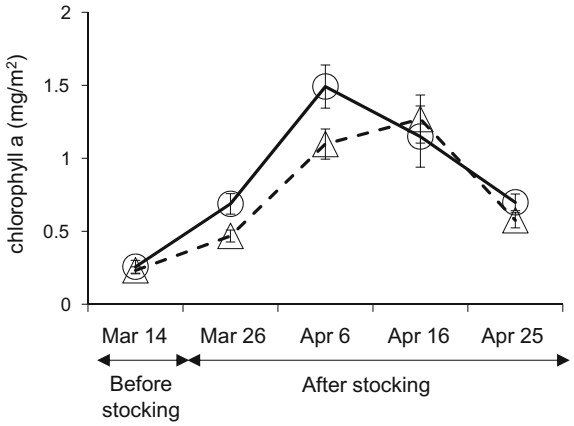

Fig. 5 Mean values (±SE) of chlorophyll-*a* content from algae recovered from eight experimental terracotta tiles placed at the study sites, in stream sections with (sites 1–4) and without (sites 5–8) hatchery chum salmon fry. Sites 1–4 and 5–8 are represented by *circles* joined by *solid lines* and *triangles* joined by *dashed lines*, respectively

Table 5 Summary of generalized linear mixed model for chlorophyll *a* density

	df1	df2	*F*	*P*
Date	4	182	51.68	<0.001
Presence/absence of hatchery chum salmon fry (P/A)	1	83	3.605	0.061
Date × P/A	4	182	2.540	0.041

Our results suggest that a single release of hatchery chum salmon fry into a stream may influence the stream ecosystem. Hatchery chum salmon fry disappear from the stream due to seaward migration (e.g., Hasegawa & Takahashi, 2013). Therefore, their effects on the stream ecosystem may be limited. For example, stomach fullness and body weight of masu salmon with hatchery chum salmon fry recovered gradually and the difference in algal biomass between study sites with and without hatchery chum salmon fry vanished. Conversely, abundance of some invertebrate categories, such as total grazers, were different between study sites with and without hatchery chum salmon fry during the survey. Further studies are required to accurately determine how long the effects of hatchery chum salmon last on the stream ecosystem after stocking.

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